

# **World-views, Preconceptions, Biases and How to Change your Mind**

**ENGR49580D Technology, Society and Sustainability**

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**Once facts are established,  
opinions can be formed.  
And while opinions matter,  
they don't change the facts.**

**-CNN** ([https://www.cnncreativemarketing.com/project/cnn\\_factsfirst/](https://www.cnncreativemarketing.com/project/cnn_factsfirst/))

# Cognitive Biases

- People's world-views begin to form in childhood and are reenforced over time by peers and media consumption
- An attack on an individual's world-view may be seen by that individual as an attack on their identity

# Cognitive Biases

## Questioning Reality

- When encountered with evidence that contradicts their world-view, an individual is faced with two options:
  - Change their mind
    - This statement may be better phrased as the more literal *change their brain*, as the formation of new neural pathways may be involved
  - Reject incompatible evidence

# Cognitive Biases

## Questioning Reality

- Confronting facts that contradict an individual's world-view may serve to strengthen that individual's world view rather than change it
- This is known as belief perseverance

# Cognitive Biases

## Confirmation Bias

- The natural tendency to seek out information that supports your beliefs
  - Seeking out like minded people and media
  - Reduces the ability to analyze facts objectively

# Brain Biology

## Dopamine, Adrenaline and Cortisol

- Winning a debate can release numerous hormones including dopamine and adrenaline
- Situations of high stress or distrust can release cortisol
  - Cortisol is a 'stress hormone' that prepares the body for immediate survival situations (the 'fight or flight' reaction)
    - Cortisol shuts down the executive function of the brain (logic and reason) in favour of faster decision making processes

# How to Change your Mind

- Work actively to keep an open mind
  - Seek out new perspectives on issues
- Don't be swayed by outliers
- Be wary of repetition
  - Repeated statements are perceived as more truthful than new ones
- If you feel your body reacting, pause and gather yourself

# How to Change Other's Minds

- Listen Actively
- Present things in a non-confrontational way
  - Don't make others feel attacked
- Do not insult (misinformed is an insult)
- People are more likely to change their minds because of a personal connection than a call to authority
- Accept that you will probably fail

# **The 'Life Cycle' Perspective**

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# Life-Cycle Assessment

## Overview

- Studies the environmental impact of a product through the entire product lifespan from material extraction to end of life disposal
- Considers issues beyond day to day product use (and cost) including:
  - embedded energy / carbon
  - expected useful life
  - end of life impacts

# Life-Cycle Assessment

## A formal study methodology

- ISO defines a method for Life-Cycle Assessment (LCA) of environmental impacts of a product
- LCA is a data driven process based on:
  - The materials used on the product as well as the known impacts of those materials
  - The processes used to transform those material and the known impacts of those processes

# Life-Cycle Assessment

## Limitations to the methodology

- LCA relies on a defined system boundary changing the boundary can change the results
- LCA relies on data sets, if specific data is not available, industry standard data is substituted
- Many steps in the LCA process rely on individual judgement
  - Different assessments of the same product may lead to different results

# Life-Cycle Assessment

## Types of LCA

- Environmental Product Declarations
- Studies compliant with a product-specific (or sector-specific) standard
- Single-issues analyses like the carbon or water footprint
- Social LCA
- Organizational LCA
- Long-term monitoring studies

# Life-Cycle Assessment

## Life-Cycle Variants

- Cradle to Grave (traditional Life-Cycle)
- Cradle to Gate
  - Analysis ends at completion of manufacturing
  - Does not consider end use or disposal
- Well to Wheel
  - Used to assess the impact of transport fuels

# Life-Cycle Assessment

## Life-Cycle Variants

- Cradle to Cradle
  - targets complete recycling of the end of life product as well as use byproducts
  - aspires towards a circular economy with minimal (0) inputs or outputs
- Environmental Impact Assessment
  - Used for infrastructure construction projects

# **The Role of Public Policy**

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# **The Role of Public Policy**

## **A conflict between divergent interests**

- Corporations are responsible to their share-holders to turn a profit
  - In many cases this is an actual legal responsibility

# **The Role of Public Policy**

## **A conflict between divergent interests**

- Many process (especially in resource extraction and manufacturing) generate impacts that go beyond the corporation undertaking those process
- These may include:
  - The discharge of solid, liquid or gaseous waste
  - The depletion of mineral resources
  - The destruction of natural systems
- These are called externalities because their cost is not born by the party benefiting

# **The Role of Public Policy**

## **A conflict between divergent interests**

- The public has an interest in limiting externalities in order to:
  - Preserve resources for current and future generations
  - Prevent degradation of health due to pollution
  - Allow land to continue to be usable for other purposes such as forestry, agriculture, recreation

# **The Role of Public Policy**

## **A conflict between divergent interests**

- The public also has an interest in manufacturing and resource extraction processes occurring
  - Food production
  - Employment
  - Consumer Goods
  - Energy

# **The Role of Public Policy**

## **Tools to balance divergent interests**

- The role of government is (supposed to be) to balance public needs
- Public policy tools include:
  - Taxation (such as carbon taxes)
  - Regulation (limits on certain types of discharge, producer responsibility)
  - Performance Bonds (to ensure clean up / reclamation)
  - Infrastructure (to centrally mitigate impacts)